An age-structured contact model of sick-leave and school-absence effects on seasonal influenza in Korea

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Abstract

Background. Keeping infectious children out of school is a comparatively well-studied influenza control measure, but the effect of keeping infectious adults home from work (sick leave) is far less established, and the two have rarely been evaluated on the same footing. Comparing them for an age-structured pathogen requires attention to three things a single effect-size number misses: transmission removed from one setting (workplace or school) is not eliminated but redistributed to others, particularly the household; a substantial fraction of onward transmission is presymptomatic and therefore escapes any symptom-triggered absenteeism; and the effect can differ across age groups and with the timing of the intervention relative to the school calendar.

Methods. We built an age-structured deterministic SVEIR model for the Seoul Capital Area with 15 age groups, an Erlang-distributed infectious period ($I_1 I_2 I_3$) that separates a presymptomatic stage (I_1) from the symptomatic stages (I_2, I_3), and four contact channels (home, work, school, other), calibrated to National Health Insurance Service (NHIS) claims data across three normal seasons (2016–17, 2017–18, 2019–20). School-calendar changes in contact frequency were captured by a time-switching contact matrix; intervention intensity by a within-home crowding term acting on symptomatic absentees; and presymptomatic transmission by a stage-specific infectiousness weight. Channel allocation and season-specific reproduction numbers were estimated by a three-season joint No-U-Turn Sampler (NUTS) run (4000 post-warmup draws, zero divergences), and sick-leave and school-absence policies were compared at matched intensity by posterior forward simulation with credible intervals.

Results. First, school absence was both more effective and more certain than sick leave: it averted +1.56 to +3.36% of infections with a credible reduction in every age band and season (all $0 \notin$ CI) — roughly fivefold more than sick leave in absolute counts (2019–20: 123,147 vs 23,492 infections averted) — because the school channel, small in transmission share ($\pi_{\text{school}} \approx 0.065$) but child-dominated, is the epidemic's engine; sick leave's net benefit was small (+0.23 to +0.46%) and its credible interval included zero in two of three seasons. Second, sick leave redistributes rather than simply reduces: adult infection fell while school-age children's rose, a direction reproduced in all three independently fitted seasons but small enough to

be only partly significant (a redistribution tendency, not established harm to children), and the direction differed by metric — children in attack-rate terms, adults in absolute counts. Third, the redistribution toward children was larger during the winter break than during the school term, because sick-leave spillover flows through the always-open home channel; the direction was consistent but its magnitude was sensitive to immunity and crowding assumptions, so we describe it as redistribution that intensifies during the winter break rather than a clean sign reversal. The joint calibration also identified the workplace transmission share, $\pi_{\text{work}} = 0.298$ (95% CI $[0.241, 0.356]$), which single-season fits cannot resolve.
Conclusions. For an age-structured pathogen with presymptomatic transmission, school-based measures are both more effective and more certain than sick leave, and are free of the household-spillover hazard. Sick leave's population-level benefit is small and not robustly established; where used, it should be paired with household-isolation support, especially during winter holidays when its redistribution toward children is largest. Policy appraisal should weigh whom a measure moves risk toward, and when it is applied, rather than its aggregate effect size alone.

Keywords: seasonal influenza; age-structured transmission; sick leave; school absenteeism; household spillover; presymptomatic transmission; non-pharmaceutical interventions

1 Introduction

Keeping infectious individuals out of a shared setting — paid sick leave for symptomatic workers, and school exclusion for symptomatic children — is a standard non-pharmaceutical response to seasonal and pandemic influenza. The intuitive case is that an infectious person who stays home cannot infect the colleagues or classmates they would otherwise have met, so removing them lowers total infection. Evaluations have accordingly concentrated on the size of that reduction.

For an age-structured respiratory pathogen this framing omits four things. First, the population is not homogeneous: susceptibility, clinical attack rate, and contact patterns vary sharply by age, and children are both among the most susceptible and, in most seasons, the most infectious group [12, 2]. A case prevented among adults is not equivalent to a case prevented among children. Second, sending an infectious person home does not end their transmission; it relocates it into the household, whose age composition differs entirely from the setting they left. An infectious adult kept from work exposes, above all, their own children. Third, symptom-triggered absenteeism cannot act on transmission that occurs before symptoms: a meaningful share of influenza A transmission is presymptomatic [4, 13], and that share escapes any policy keyed to staying home when sick. Fourth — and least appreciated — when the policy is applied matters, because the household is always open even when schools are not. A measure that merely shifts transmission into the home behaves differently during the school term, when children are dispersed at school, than during a holiday, when they are at home all day.

These features interact in ways an effect-size analysis cannot see. Sick leave takes an infectious adult out of an adult-dominated setting and relocates their residual, post-symptomatic transmission into the household, toward children; school absence takes an infectious child out of a child-dominated setting and relocates transmission toward co-resident adults. The two levers are not two instances of one "reduce contact" mechanism but act on different channels and different ages. And because the sick-leave pathway runs through the home rather than the school, its burden on children depends on how much time children spend at home — which the school calendar governs. Figure 1 sketches the two pathways.

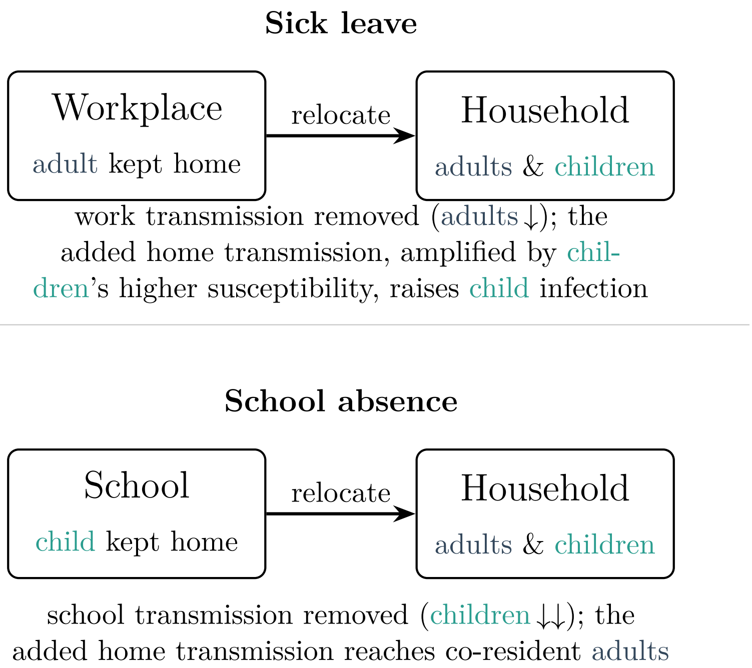

Figure 1: The two absenteeism levers as transmission-redistribution operators. Each policy keeps an infectious individual home from one setting (workplace or school) and adds their residual post-symptomatic transmission to the household, whose members span all ages. Because the removed source and the household's age mix differ, the net effect is a shift along the age gradient: sick leave moves risk from adults toward co-resident children, and school absence moves a smaller amount from children toward co-resident adults. The household spillover is not a fixed adult-to-child (or child-to-adult) route; it is a change in who bears the household's increased transmission. In the home contact matrix a child's contacts are predominantly with adults (≈80%) and an adult's are also predominantly with other adults (≈78%); school absence therefore relocates a sick child's transmission chiefly onto co-resident adults, whereas sick leave's net shift onto children reflects children's higher susceptibility (and their lack of any offsetting workplace reduction) more than raw contact volume. Colours denote age groups (adults, children); boxes denote settings.

We argue that the outcome of an absenteeism policy is determined by dimensions the effect-size literature collapses: the direction in which it redistributes transmission along the age gradient of vulnerability, the metric used to read that redistribution, and the timing of the intervention relative to the epidemic and the school calendar. This reframes the evaluation question from "how much does the policy avert" to "whom does it move the risk toward, with what certainty, and when."

To examine this we developed an age-structured SVEIR model of seasonal influenza in the Seoul Capital Area, with an Erlang-distributed infectious period that isolates a presymptomatic stage, four explicitly separated contact channels, and a physically grounded representation of what absenteeism does to contacts, calibrated to three normal seasons of NHIS claims data and equipped with full posterior credible intervals from a three-season joint NUTS calibration. The model lets us (i) compare sick-leave and school-absence policies at matched intensity on the same footing and with credible intervals; (ii) locate where in the workplace-transmission / crowding parameter space sick leave helps versus harms; (iii) decompose outcomes by age and by metric; and (iv) ask how the same sick-leave policy behaves in different phases of the epidemic, split naturally by the winter-break calendar.

We report four findings. School absence is confidently effective across all seasons and age groups, whereas sick leave's net benefit is small and, in two of three seasons, not distinguishable from zero. School absence averts far more infection than sick leave — roughly fivefold in counts — because the school channel drives the epidemic despite its small fitted transmission share. Sick leave redistributes rather than reduces, moving infection from adults toward school-age children in direction across all three seasons, though with a magnitude small enough that the redistribution is only partly significant and should be read as a tendency rather than established harm. And that redistribution toward children is larger during the winter break than during the school term, because its spillover flows through the always-open home channel. The unifying message is that absenteeism policy for age-structured pathogens should be judged by who bears the redistributed risk, with what certainty, and when the policy is applied — not by an aggregate effect size.

2 Model Description

This section defines the model structure — compartments, flows, force of infection, and the mechanisms for presymptomatic transmission, household spillover, the school calendar, and vaccination — in symbolic form. Numerical values of every parameter are collected in Section 3, and the procedures used to calibrate and exercise the model in Section 4.

2.1 Compartmental structure and flow

We model seasonal influenza in the Seoul Capital Area as an age-structured SVEIR (susceptible–vaccinated–exposed–infectious–recovered) system stratified into 15 five-year age groups (0–4, 5–9, ..., 65–69, 70+). For each age group a, susceptibles S_a acquire vaccine-derived leaky protection (moving to V_a, efficacy VE) at the time-dependent vaccination hazard $v(t)$, or become exposed E_a at the force of infection $\lambda_a(t)$; vaccinated individuals may still be infected, at reduced hazard $(1 - \mathrm{VE})\lambda_a$. Exposed individuals become infectious at rate σ. The infectious period is Erlang-distributed with three sub-stages $I_{1,a} \to I_{2,a} \to I_{3,a}$, each left at rate 3γ (so the mean infectious period is $1/\gamma$); the shape follows Wearing et al. [22] and, crucially here, gives the model a distinct presymptomatic sub-stage I_1 (Section 2.3). The flow is shown in Figure 2, and the governing equations for each age group a are

$$\begin{aligned} \frac{dS_a}{dt} &= -\lambda_a S_a - v(t)\, S_a, & \frac{dV_a}{dt} &= v(t)\, S_a - (1-\mathrm{VE})\,\lambda_a V_a, \\ \frac{dE_a}{dt} &= \lambda_a S_a + (1-\mathrm{VE})\,\lambda_a V_a - \sigma E_a, & \frac{dI_{1,a}}{dt} &= \sigma E_a - 3\gamma\, I_{1,a}, \\ \frac{dI_{2,a}}{dt} &= 3\gamma\, I_{1,a} - 3\gamma\, I_{2,a}, & \frac{dI_{3,a}}{dt} &= 3\gamma\, I_{2,a} - 3\gamma\, I_{3,a}, \\ \frac{dR_a}{dt} &= 3\gamma\, I_{3,a}. \end{aligned} \tag{1}$$

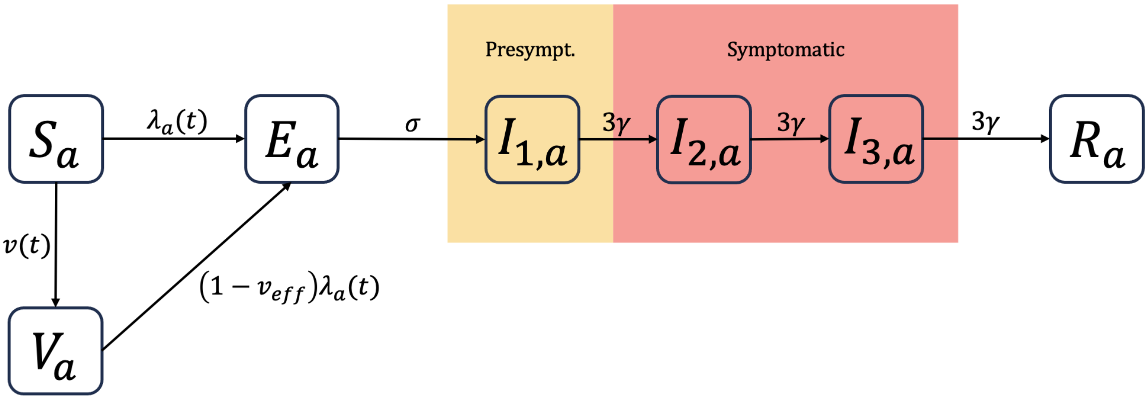

Figure 2: Compartmental flow for each age group a. The infectious period is an Erlang(3) chain; I_1 is presymptomatic (infectious at reduced weight w, not subject to absenteeism), I_2 and I_3 are symptomatic (subject to policy). Symptom onset, and the observation model, are placed at the $I_1 \to I_2$ transition. Here $v_{\mathrm{eff}} = \mathrm{VE}$ is the vaccine efficacy.

2.2 Force of infection and contact channels

The force of infection is separated into four contact channels $c \in \{\mathrm{home}, \mathrm{work}, \mathrm{school}, \mathrm{other}\}$. Writing $I_{1,b}$ and $I_{23,b} = I_{2,b} + I_{3,b}$ for the presymptomatic and symptomatic prevalences in age group b, the channel hazards are

$$\lambda_a^{\mathrm{work}} = \phi_a\, s_f(t)\, \beta_{\mathrm{work}} \sum_b C_{ab}^{\mathrm{work}}(t)\, \frac{w\, I_{1,b} + p_{\mathrm{work}}\, I_{23,b}}{N_b}, \tag{2}$$

$$\lambda_a^{\mathrm{school}} = \phi_a\, s_f(t)\, \beta_{\mathrm{school}} \sum_b C_{ab}^{\mathrm{school}}(t)\, \frac{w\, I_{1,b} + p_{\mathrm{school}}\, I_{23,b}}{N_b}, \tag{3}$$

$$\lambda_a^{\mathrm{home}} = \phi_a\, s_f(t)\, \beta_{\mathrm{home}} \sum_b C_{ab}^{\mathrm{home}}(t)\, \frac{w\, I_{1,b} + \big(1 + \kappa_b(1-p)_b\big) I_{23,b}}{N_b}, \tag{4}$$

$$\lambda_a^{\mathrm{other}} = \phi_a\, s_f(t)\, \beta_{\mathrm{other}} \sum_b C_{ab}^{\mathrm{other}}(t)\, \frac{w\, I_{1,b} + I_{23,b}}{N_b}. \tag{5}$$

with total hazard $\lambda_a(t) = \sum_c \lambda_a^c(t)$. Here ϕ_a is age-specific relative susceptibility, $s_f(t)$ a seasonal forcing term, β_c the transmissibility weight of channel c, and $C_{ab}^c(t)$ the time-dependent contact matrix of channel c between age groups a and b (age-structured contact matrices follow the POLYMOD approach [16]). The presymptomatic block $w\, I_1$

passes through unaffected in every channel; the policy retention fractions $p_{\text{work}}, p_{\text{school}}$ (Section 2.4) and the home crowding term $\kappa_b(1-p)_b$ act only on the symptomatic block I_{23}.

2.3 Presymptomatic transmission and the observation model

The first infectious sub-stage I_1 represents the presymptomatic phase: individuals are infectious but not yet symptomatic. Symptoms — and hence any symptom-triggered behaviour (care-seeking, absenteeism) — begin at the $I_1 \to I_2$ transition, so we place the observation model (NHIS claims) at symptom onset and let both policies act only on the symptomatic stages I_2, I_3. Presymptomatic individuals transmit at a reduced relative infectiousness w; every channel's force of infection therefore weights I_1 by w and I_2+I_3 by unity. The value of w (Section 3) follows a recent household study of influenza A [4] and earlier shedding estimates [13]; influenza B shows no clear evidence of presymptomatic transmission, so applying an A-based weight throughout is a conservative choice.

Because absenteeism cannot touch presymptomatic transmission, the presymptomatic stage sets a ceiling on what any symptom-based policy can achieve, since some of the transmission a sick worker generates occurs through I_1 contacts the policy does not reach. This ceiling applies equally to any symptom-triggered policy; as the results below show, sick leave's small net effect is driven mainly by household spillover rather than by this presymptomatic ceiling.

2.4 Household spillover under absenteeism

Absenteeism is represented as additional within-home transmission incurred when symptomatic individuals who would otherwise have attended work or school instead remain at home. Baseline attendance is described by a retention fraction p (the fraction who keep attending while symptomatic), so $1-p$ is the fraction sent home; the policy raises the intervention intensity by lowering p (Section 4.4). Only the symptomatic block I_{23} is affected — presymptomatic individuals are, by definition, not yet identifiable as sick — and the home-channel symptomatic prevalence is scaled by

$$1 + \kappa_b\,(1-p)_b, \tag{6}$$

where κ_b is the per-group increase in effective household transmission from a symptomatic individual spending the day at home, and $(1-p)_b$ is the group-specific sent-home fraction:

$$(1-p)_b = \begin{cases} 1-p_{\text{school}}, & \text{students,} \\ \rho_b\,(1-p_{\text{work}}), & \text{workers,} \\ 0, & \text{70+,} \end{cases} \tag{7}$$

with ρ_b the measured employment rate (students take $\rho_b = 0$, i.e. no commute). We deliberately write the spillover through $1-p$ rather than introducing a separate "retained-fraction" symbol: the retention p already carries this information, and the crowding coefficient κ is the only genuinely new quantity. The crowding coefficients are derived independently — not fitted — from a national time-use survey, and their construction and values are given in Section 3.

2.5 Time-switching contact matrix $C(t)$

The school calendar enters the model through the contact matrix, not the transmissibility coefficient. Each channel's matrix is a trapezoidal-in-time blend of an empirically measured school-term matrix and winter-break matrix,

$$C^{c}_{ab}(t) = \Big(1 - h(t)\Big)\, C^{c,\text{term}}_{ab} + h(t)\, C^{c,\text{vac}}_{ab}, \tag{8}$$

where C^{term} and C^{vac} are the school-term and winter-break contact matrices from a nationwide Korean contact survey conducted by the National Institute for Mathematical Sciences [20, 3], consistent with the marked holiday reduction in school-age contacts reported internationally [7], and $h(t)$ is the winter-break calendar (a trapezoidal ramp; dates in Section 3); all four channels switch together. This matters because an earlier version encoded winter break as a multiplier on β_{school}, which conflates contact number with transmissibility per contact and spuriously inflated the baseline home force of infection during the winter break. Moving the calendar into $C(t)$ makes the baseline spillover factor identically 1.0 across the year (verified numerically), so that contact frequency is handled by $C(t)$ and intervention intensity by κ, and the two never contaminate each other.

2.6 Vaccination

Vaccine uptake enters as a time-distributed, age-specific hazard normalized to reach an age-specific target coverage C_a over the finite season,

$$v_a(t) = \frac{-\ln(1 - C_a)}{Z}\, \text{density}(t), \tag{9}$$

where $-\ln(1 - C_a)$ converts target coverage to cumulative hazard, density(t) is the seasonal uptake shape (shared across ages), and Z normalizes it over $[0, 365]$. The age-specific coverages C_a (Table 2) follow Korea's free-vaccination program, which prioritizes young children and the elderly. This hazard transform reproduces the target vaccinated fraction essentially exactly, correcting an earlier density-scaling implementation that undershot coverage.

2.7 Transmissibility allocation

To separate the scale of transmission from its allocation across channels, we write

$$\beta_c = \frac{R_0}{\rho_{\text{eff}}(\pi, \phi)}\, \pi_c, \qquad \textstyle\sum_c \pi_c = 1, \tag{10}$$

where R_0 sets the size (data-determined) and $\pi = (\pi_{\text{home}}, \pi_{\text{work}}, \pi_{\text{school}}, \pi_{\text{other}})$ sets the allocation. Here ρ_{eff} is the dominant eigenvalue of the next-generation operator, adjusted for presymptomatic infectiousness: because I_1 transmits at reduced weight w, the effective spectral scaling is $\rho_{\text{eff}} = \rho \cdot (w + 2)/3$, and using ρ_{eff} keeps β and R_0 mutually consistent. The allocation π is informed by channel-specific priors and, decisively, by multi-season data (Section 4); prior (pin) strengths are listed in Table 3.

3 Parameters

This section collects the values that parametrize the model: fixed constants (Table 1), the age-specific vectors (Table 2), and the estimated parameters with their priors (Table 3). Procedural settings — data sources, initial conditions, solver, and calibration configuration — are given in Section 4.

Table 1: Fixed model constants.

Symbol	Description	Value	Source / note
$1/\sigma$	Latent period	2.0 d	Standard influenza value
$1/\gamma$	Infectious period (mean)	4.0 d	Standard influenza value
n	Erlang infectious sub-stages	3 (sub-stage rate 3γ)	[22]
w	Relative presymptomatic infectiousness	0.22	[4]: influenza A presymptomatic share 9.6% (5.9–14.7)
$(w{+}2)/3$	NGM correction factor	0.740	Keeps $\beta \leftrightarrow R_0$ under I_1 attenuation
VE	Vaccine efficacy	0.5	Standard seasonal-influenza value
$s_f(t)$	Seasonal forcing	$1 + 0.9\cos\left(\frac{2\pi(t-105)}{365}\right)$	Cosine; peak mid-December
$v_a(t)$	Vaccine-uptake hazard	$\frac{-\ln(1-C_a)}{Z}\,\mathrm{density}(t)$	Gaussian density (peak ISO wk 42, σ=4 wk); C_a age-specific (Table 2), Z≈0.9332

Note. The infectious period is an Erlang(3) chain $E \to I_1 \to I_2 \to I_3 \to R$: I_1 is presymptomatic (infectiousness scaled by w, not subject to policy) and I_2, I_3 are symptomatic (infectiousness 1, subject to policy). The NGM correction leaves R_0 unchanged.

3.1 Age-specific vectors

Relative susceptibility ϕ_a follows a linear U-shape anchored at a child value of 2.0 [2, 14], a working-age floor of 1.0, and an elderly value of 1.5 [17], with intermediate bands interpolated linearly and consistent with age-specific attack-rate evidence [12]. Reporting fractions $\gamma_{\mathrm{report},a}$ follow CDC-style estimates [18], with the adolescent (15–19) band set to the adult level to reflect adolescents' adult-like health-care–seeking (raising its observed/model ratio from 0.45 to 0.87). Initial immune fractions $R(0)_a$ rise with age, reflecting antigenic seniority from repeated lifetime exposure [19, 10, 6]. The household crowding coefficients κ_a were derived independently — not fitted — from the 2024 Korean Time Use Survey [21] as the proportional increase in home-stay time on a non-attendance versus attendance day (we do not further discount for co-presence, and sweep a multiplicative scale μ on κ in sensitivity analysis); they are smaller than the household-contact increases implied by Ferguson et al. [8] (+50 to +100%), as expected for seasonal influenza.

Table 2: Age-specific parameter vectors (15 five-year age bands, fixed).

Age band	ϕ_a (relative susceptibility)	$\gamma_{\text{report},a}$ (reporting fraction)	$R(0)_a$ (initial immune fraction)	κ_a (home crowding coefficient)	C_a (vaccine coverage)	ρ_a (employment rate)
0–4	2.00	0.40	0.10	0.34	0.75	0
5–9	1.75	0.40	0.10	0.34	0.75	0
10–14	1.50	0.25	0.10	0.34	0.40	0
15–19	1.25	0.18	0.10	0.34	0.40	0
20–24	1.00	0.18	0.40	0.40	0.30	
25–29	1.00	0.18	0.40	0.40	0.30	
30–34	1.00	0.18	0.40	0.40	0.30	
35–39	1.00	0.18	0.40	0.40	0.30	
40–44	1.00	0.18	0.40	0.40	0.30	measured†, $0 \to \approx 0.74$
45–49	1.08	0.18	0.60	0.40	0.30	
50–54	1.17	0.18	0.60	0.40	0.30	
55–59	1.25	0.18	0.60	0.40	0.30	
60–64	1.33	0.18	0.60	0.40	0.30	
65–69	1.42	0.25	0.65	0.40	0.30	measured†
70+	1.50	0.25	0.65	0	0.82	0

Note. ϕ_a: three literature anchors (0–4 = 2.0; 20–44 = 1.0; 70+ = 1.5) linearly interpolated [2]. $\gamma_{\text{report},a}$: CDC-style [18], 15–19 set to adult level. κ_a: student 0.34 (ages 0–19), adult 0.40 (ages 20–69), $\kappa_{70+} = 0$; co-presence discount excluded. C_a: age-specific vaccine coverage reflecting Korea's free-vaccination program (young children 0.75, adolescents 0.40, working-age 0.30, 70+ 0.82; provisional). †ρ_a (employment rate, Statistics Korea) is 0 for student bands and rises to ≈ 0.74; students take $\rho_a = 0$ (no commute), so κ acts on the home channel only.

3.2 Estimated parameters

Three quantities are estimated (Table 3): the channel allocation π (a 4-channel simplex shared across seasons), the season-specific reproduction numbers $R_{0,s}$, and a negative-binomial observation dispersion ϕ_{NB}. The allocation carries channel-specific log-normal "pins" on logit π_c, centred on a reference allocation with $\pi_{\text{work}} = 0.29$ [20]: the school pin is deliberately strong to prevent seasonality or child effects from being absorbed into that channel, while the work pin is left relaxed ($\sigma_{\text{pin}} = 0.10$) so that the multi-season data, not the prior, fix π_{work} (Section 4.3). The reproduction numbers take a weakly informative prior $\log R_0 \sim \mathcal{N}(\log 2.0,\ 0.5)$ and the dispersion $\log \phi_{\text{NB}} \sim \mathcal{N}(\log 10,\ 1.5)$. The sampler configuration and convergence diagnostics are in Section 4.2.

Table 3: Estimated parameters: priors and posteriors (mean [95% CI]).

Parameter	Pin σ_{pin}	Posterior mean [95% CI]
Channel allocation π_c		
π_{home}	0.15	0.442 [0.371, 0.513]
π_{work}	0.10	0.298 [0.241, 0.356]
π_{school}	0.05	0.065 [0.056, 0.074]
π_{other}	0.15	0.195 [0.158, 0.236]
Reproduction number $R_{0,s}$		
2016–17	—	2.287 [2.208, 2.372]
2017–18	—	2.078 [2.008, 2.152]
2019–20	—	2.169 [2.098, 2.250]
Observation dispersion ϕ_{NB}	—	1.43 [1.26, 1.61]

4 Methods

This section describes how the model of Section 2, with the parameters of Section 3, was calibrated to data and used to compare policies.

4.1 Data and seasons

The model is calibrated to influenza incidence from National Health Insurance Service (NHIS) claims for the Seoul Capital Area, obtained through the NHIS National Health Insurance Sharing Service (NHISS) research data platform (`https://nhiss.nhis.or.kr`); influenza cases were identified from claims records and aligned to symptom onset (the $I_1 \to I_2$ transition). We use three "normal" seasons — 2016–17, 2017–18, and 2019–20 — restricting attention, in each season, to the first (dominant) epidemic wave. Seasons with atypical dynamics were excluded: 2015–16 (slow, atypical onset) and 2018–19 and 2022–23 (double-peaked), which lie outside the scope of a single-wave model. Two further inputs are used: age-structured contact matrices for the school term and winter break, from the NIMS contact survey [20, 3], and season-specific resident-registration populations (Statistics Korea, KOSIS; June of each peak year, summed over Seoul, Incheon, and Gyeonggi). The model is run for a single aggregated Seoul Capital Area. Contact-matrix dimensions, the winter-break transition schedule, and the population-by-district source table are provided in the Supplement.

4.2 Bayesian calibration

Fitting uses a JAX / diffrax / NumPyro implementation, with posterior sampling by the No-U-Turn Sampler [11]. Each season is fitted, and a joint calibration shares a single channel allocation π across all three seasons while letting season-specific scale parameters (R_0 and reporting overdispersion) vary, to test whether the allocation is a multi-year constant. The joint posterior comes from a three-season joint NUTS run: 500 warm-up + 500 samples × 4 chains, replicated with different RNG seeds from the same post-warm-up state, for a total of 4000 post-warm-up draws (max tree depth 8, target acceptance 0.9). All channel-mix parameters π_c reached $\hat{R} \leq 1.05$ with $\mathrm{ESS}_{\mathrm{bulk}} \geq 144$; π_{work}, the least-constrained channel, had $\hat{R} = 1.016$ and $\mathrm{ESS}_{\mathrm{bulk}} = 373$. Season-specific R_0 had $\hat{R} \leq 1.06$ with $\mathrm{ESS}_{\mathrm{bulk}} \geq 118$, and there were zero divergent transitions. Initial conditions were set per season: $I(0)$ from NHIS incidence over the first three weeks divided by γ_{report} (distributed evenly across the three Erlang stages), $E(0) = 0.5\,I(0)$, $R(0)_a$ from age-specific initial immunity times population (Table 2), and $V(0) = 0$ (the vaccinated compartment fills in-season via $v(t)$). The SVEIR system was integrated with the Dopri5 solver (`diffrax`) on a daily grid over $[0, 364]$ days (relative/absolute tolerance $10^{-4}/10^{-6}$), with observations taken at symptom onset. The calibrated model reproduces the observed incidence in all three seasons (Figure 3).

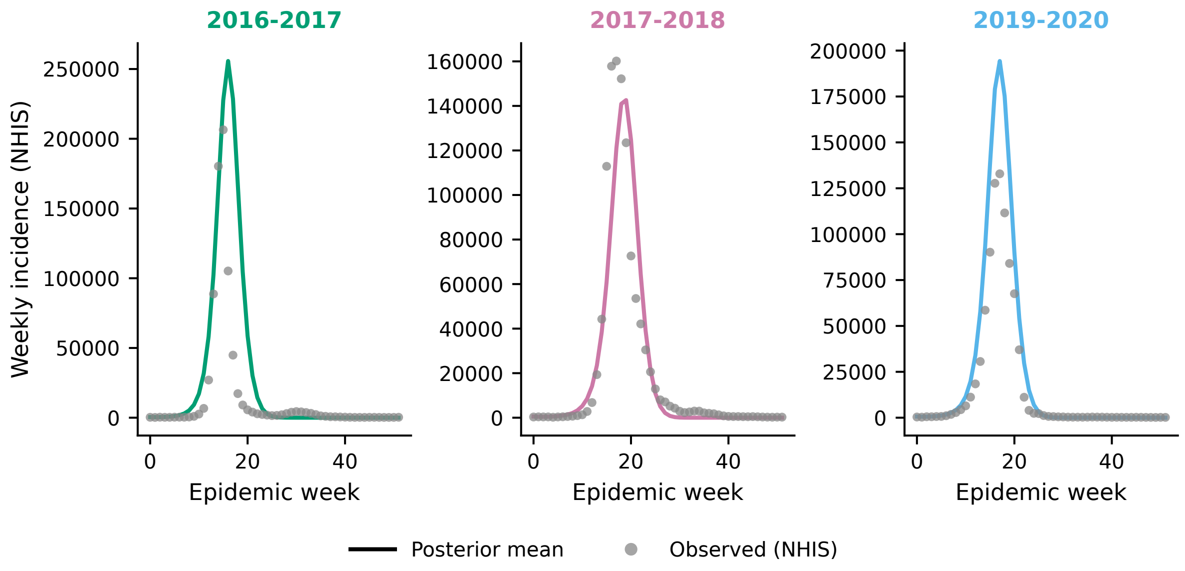

Figure 3: Model fit: observed NHIS influenza incidence (points) versus the posterior mean model (line), by season, aligned to symptom onset.

4.3 Identifiability of workplace transmission

Freely estimated from a single season, β_{work} collapses toward zero (railing in 88–94% of unconstrained fits) — workplace transmission is not identifiable from one season alone. This is practical, not structural, non-identifiability with a structural cause: the home and work contact matrices are similar in shape among working-age adults, so the two channels trade off, and raising β_{work} can be offset by the presymptomatic weight w with little change in likelihood. The single-season silence of β_{work} is not evidence that workplace transmission is absent; it is evidence that one season cannot separate it — a gap the multi-season joint calibration closes (Section 5.4). We therefore retain an informative work pin near literature values (Table 3) and report full sensitivity across the plausible range.

4.4 Policy operationalization and epidemic-phase windows

Policies are applied as time-windowed reductions in the retention fraction,

$$p_{\text{eff}}(t) = 1 - w_{\text{pol}}(t)\,(1-p), \tag{11}$$

where $w_{\text{pol}}(t)$ is the policy window and p the residual retention (intensity $1-p$); the school and work operators act on independent windows and channels, and both act only on the symptomatic block I_{23}. Comparisons between policies use posterior forward simulation over $N = 500$ posterior draws per season at the joint allocation $\pi =$ [home 0.442, work 0.298, school 0.065, other 0.195], with $\beta_c = (R_0/\rho_{\text{eff}})\,\pi_c$ evaluated at each season's posterior R_0; we sweep intensity $p \in \{1.0, 0.8, 0.6, 0.4, 0.2\}$ and crowding scale $\mu \in \{0.5, 1.0, 1.5\}$.

Because winter influenza overlaps the winter break, the epidemic period splits naturally by the winter-break calendar into a school-term window and a winter-break window. These windows differ intrinsically in length and in background transmission, so we deliberately avoid comparing total averted-infection magnitudes across windows. Instead we report window-length-robust quantities: relative effect between policies within the same window, age-redistribution direction (independent of window length), and the direction and relative size of a policy's effect on a given age group across windows.

4.5 Sensitivity analysis

Because several structural inputs carry uncertainty, we test the robustness of the conclusions — especially the age-redistribution direction and its behaviour across the school term and the winter break — by re-running the forward comparison over sweeps of the presymptomatic weight w, the crowding scale μ, the workplace share π_{work}, adult initial immunity, the symmetric baseline retention p, and the number of Erlang sub-stages. These reuse the posterior draws (no re-fitting) and are reported stratified by age group and by term/winter-break window, so that the qualitative conclusions — rather than only the total effect size — are what is stress-tested.

5 Results

5.1 School absence is confidently effective; sick leave's net benefit is small and uncertain

The clearest and most robust finding concerns certainty, not just magnitude. Under the posterior forward simulation, school absence produced a credible reduction in total infection in every season, with averted infection of +1.56% (90% CI $[+0.91, +2.29]$) in 2016–17, +3.36% ($[+2.06, +5.02]$) in 2017–18, and +2.12% ($[+1.23, +3.19]$) in 2019–20 — all excluding zero (Table 4, Figure 4). Sick leave, by contrast, produced a much smaller net effect whose credible interval included zero in two of the three seasons: +0.46% ($[+0.03, +0.88]$) in 2016–17 but +0.23% ($[-0.37, +0.83]$) in 2017–18 and +0.40% ($[-0.08, +0.90]$) in 2019–20. The message is not that sick leave is harmful — its point estimates are positive — but that its population-level benefit is small and, under the posterior, not statistically established, whereas school absence's is.

Table 4: Total averted infection (%) by policy and season, posterior mean and 90% credible interval. $\star$ denotes $0 \notin$ CI.

Season	Sick leave	School absence
2016–17	+0.46% $[+0.03, +0.88]$ $\star$	+1.56% $[+0.91, +2.29]$ $\star$
2017–18	+0.23% $[-0.37, +0.83]$	+3.36% $[+2.06, +5.02]$ $\star$
2019–20	+0.40% $[-0.08, +0.90]$	+2.12% $[+1.23, +3.19]$ $\star$

That sick leave's net effect is limited is itself a finding, and it follows directly from a model feature grounded in data: the transmission sick leave removes from the workplace is not eliminated but relocated into the household, where it partly re-emerges as spillover. Sick leave's benefit is thus the fragile difference between transmission removed at work and transmission re-created at home. Presymptomatic transmission caps what any symptom-triggered policy can achieve, but it is not the main driver of sick leave's small net effect — the conclusion is unchanged when the presymptomatic stage is removed — whereas the household-spillover mechanism is decisive.

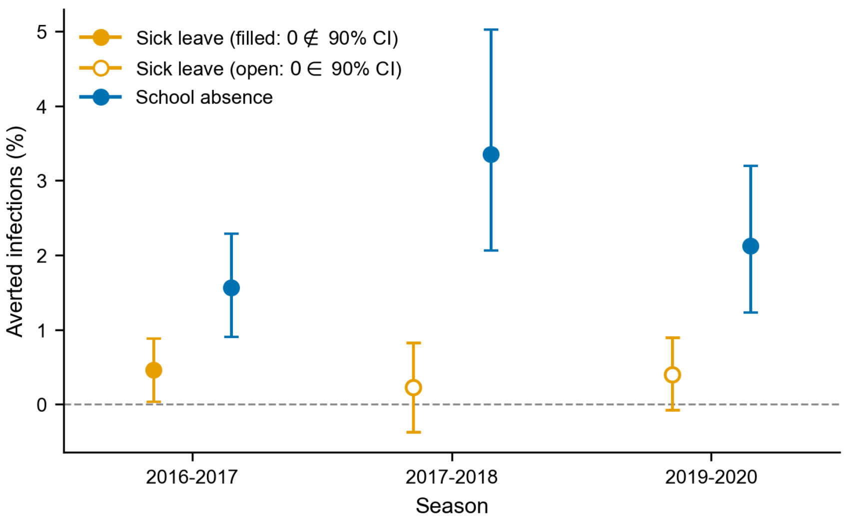

Figure 4: School absence versus sick leave: total averted infection with posterior credible intervals.

5.2 School absence averts more, and sick leave redistributes: rate versus number

Beyond certainty, school absence is also the larger lever: in absolute infection counts for the representative 2019–20 season it averted about 123,147 infections against sick leave's net 23,492 — a roughly fivefold difference (the precise ratio varies across seasons and, because sick leave's effect sits near zero, is imprecise, so we report the direction rather than a fixed multiple). The asymmetry follows from the structure of the epidemic: the school channel carries only a small share of transmission ($\pi_{\text{school}} \approx 0.065$) yet is dominated by children — among the principal transmitters and the highest-attack-rate group — so it is the engine of the epidemic and suppressing it yields disproportionate returns.

Decomposing by age reveals that sick leave redistributes rather than uniformly reduces. In all three independently fitted seasons adult attack rate fell (18–44, 45–64) while school-age children's rose (0–5, 6–11, 12–17), with the elderly essentially unaffected — the signature of a parent-to-child household pathway (Table 5). This direction (adult-down / child-up) is robust (3 of 3 seasons), but its magnitude is small and only partly significant: in 2016–17 every child band's increase excluded zero, in 2019–20 only the 6–11 band did, and in the milder 2017–18 season the child increases were positive but their intervals included zero. We therefore describe it as a redistribution tendency — consistent in direction, small in size — and do not claim that sick leave demonstrably harms children.

Which group "bears" this redistribution depends on the metric, and both readings are correct (Figure 5). In attack-rate terms (per-capita risk; Figure 5A) the effect is channel-adjacent: sick leave lowers adults and raises children, while school absence benefits children most. In absolute counts (Figure 5B), however, the adult bands dominate for both policies, because working-age adults are so numerous (the 18–44 population of the Seoul Capital Area is ∼10.0 million): in 2019–20 sick leave averted about 28,151 adult infections against about 4,349 additional child infections, for a net of +23,492. The same policy therefore looks like a per-capita transfer onto children and, at the same time, a net reduction dominated by adults — "who benefits" has no metric-free answer, and policy appraisal must state which metric it uses.

Table 5: Age-specific change in attack rate (Δattack, percentage points) under each policy, posterior mean. Sick leave shown by season; school absence as the range across the three seasons. $\star$ denotes $0 \notin$ CI for that cell; $\star\star$ denotes $0 \notin$ CI in all three seasons.

Age	Sick 16–17	Sick 17–18	Sick 19–20	School absence (3-season range)
0–5	+0.12 $\star$	+0.03	+0.07	−0.62 to −0.73 $\star\star$
6–11	+0.18 $\star$	+0.07	+0.13 $\star$	−1.24 to −1.37 $\star\star$
12–17	+0.16 $\star$	+0.06	+0.11	−1.23 to −1.29 $\star\star$
18–44	−0.23 $\star$	−0.07	−0.16 $\star$	−0.37 to −0.65 $\star\star$
45–64	−0.20 $\star$	−0.07	−0.14 $\star$	−0.24 to −0.44 $\star\star$
65+	+0.02	+0.00	+0.01	−0.10 to −0.23 $\star\star$

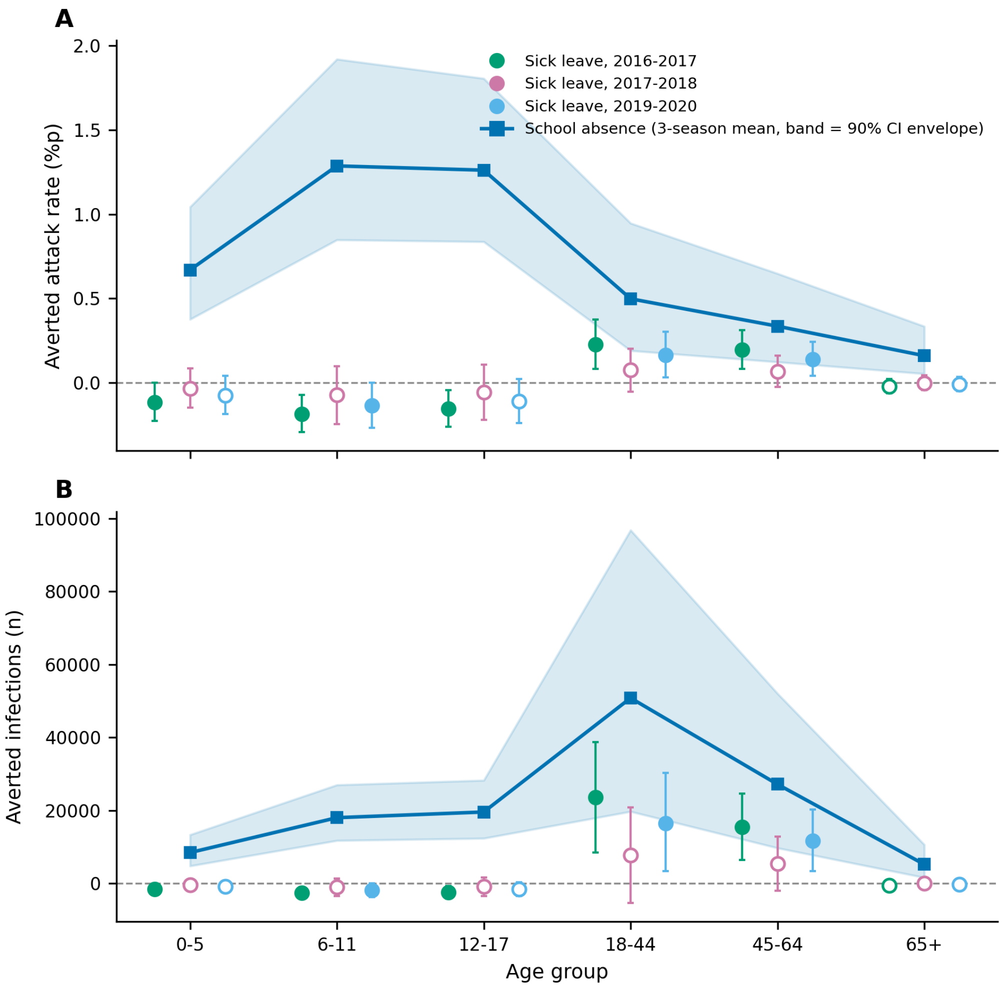

Figure 5: By-age averted effect of the two policies in two metrics: **(A)** per-capita attack rate (percentage points) and **(B)** absolute infection counts. Sick leave is shown for all three seasons (season-coloured; filled markers $0 \notin$ CI, open markers $0 \in$ CI); school absence is shown as the three-season mean (line) with a band spanning the per-season 90% credible intervals. Positive values denote averted infection. Per-capita, sick leave lowers adults and raises children while school absence benefits children most; in absolute counts, the adult bands dominate for both policies.

5.3 The redistribution toward children is larger during the winter break

Holding the sick-leave policy fixed and comparing epidemic phases, the redistribution toward children is larger during the winter-break window than during the school term (Figure 6). The mechanism is that sick-leave spillover flows through the home channel (work $\rightarrow$ home $\rightarrow$ children), whereas the winter break closes the school channel — a different pathway. The spillover route is therefore not the one the holiday shuts off; it operates regardless of whether school is in session, and during the winter break children are at home more, so the relocated transmission concentrates on them more heavily. This runs against the plausible prior that "the winter break closes schools and thereby traps and reduces spillover": spillover is a home-channel phenomenon, largely decoupled from school closure. Consistent with this, the child effect in the school-term window tracks the school-absence baseline (the school channel is open to be reduced), while the child effect in the winter-break window is roughly fixed regardless of school-absence baseline, because that channel is already shut.

We state this as a change in the magnitude of the redistribution, not as a reversal of its direction. Across the sensitivity battery (Section 5.5) the child-ward transfer during the winter break did not reverse sign — it was positive in 749 of 750 crowding-sweep posterior draws — while its size scaled with the crowding coefficient κ. The robust reading is therefore "sick leave always redistributes toward children, and the redistribution intensifies during the winter break," rather than a categorical term-benefit / winter-break harm switch. Either way the policy-relevant point stands: intervention timing, not just intensity, shapes the outcome for the vulnerable age group, and the winter-holiday period is when sick leave's household spillover weighs most on children.

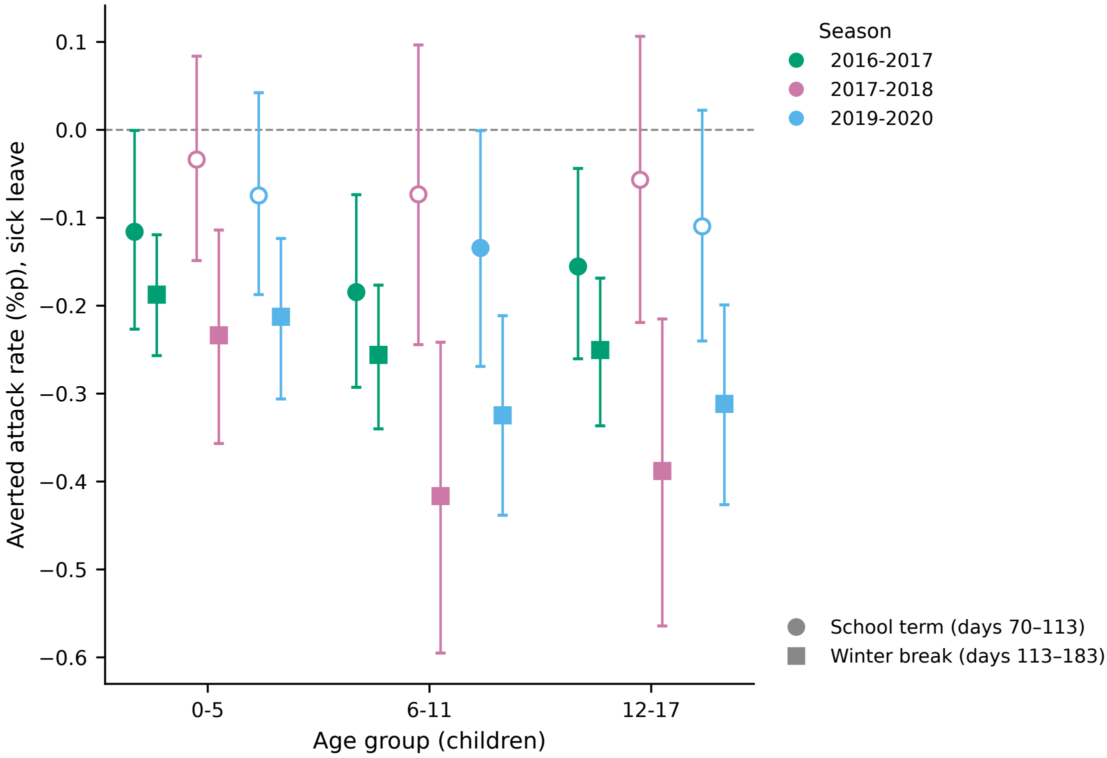

Figure 6: School term versus winter break: the sick-leave redistribution toward children intensifies during the winter break. Points are the posterior mean change in child attack rate under sick leave, compared between the school-term window (days 70–113, circles) and the winter-break window (days 113–183, squares), for each season. The vertical axis is the averted attack rate, so negative values indicate infection transferred to children. Filled markers denote a 90% credible interval excluding zero; open markers include zero.

5.4 Multi-season NUTS and identification of workplace transmission

The redistribution direction is not a one-season accident: the adult-down / school-age-up direction reproduced in 3 of 3 independently fitted seasons, with no failed fits. A single shared channel allocation reproduced all three seasons at negligible cost in fit, indicating a multi-year invariant rather than a per-season nuisance.

The three-season joint NUTS calibration also resolves the single-season non-identifiability of Section 4.3. Whereas π_{work} collapses in a single season, under the joint constraint it is identified at $\pi_{\text{work}} = 0.298$ (95% CI $[0.241, 0.356]$) — the multi-season data, not the prior, fix the value, since the pin was left at the relaxed strength $\sigma = 0.10$. The posterior channel allocation was $\pi = [\text{home } 0.442, \text{ work } 0.298, \text{ school } 0.065, \text{ other } 0.195]$, with season-specific R_0 of 2.287 (95% CI $[2.208, 2.372]$), 2.078 ($[2.008, 2.152]$), and 2.169 ($[2.098, 2.250]$) for 2016–17, 2017–18, and 2019–20 (Figure 7). The recovered π_{work} lies well above the low Italian workplace-contribution estimate (≈ 0.03; Ajelli et al. [1]) and near the domestic contact-survey estimate (≈ 0.29 [20]), consistent with the latter. Notably the data pushed home transmission well above its pin ($\pi_{\text{home}} = 0.442$ vs a reference ≈ 0.29) even with the pin relaxed — coherent with a model in which the household is the destination of sick-leave spillover.

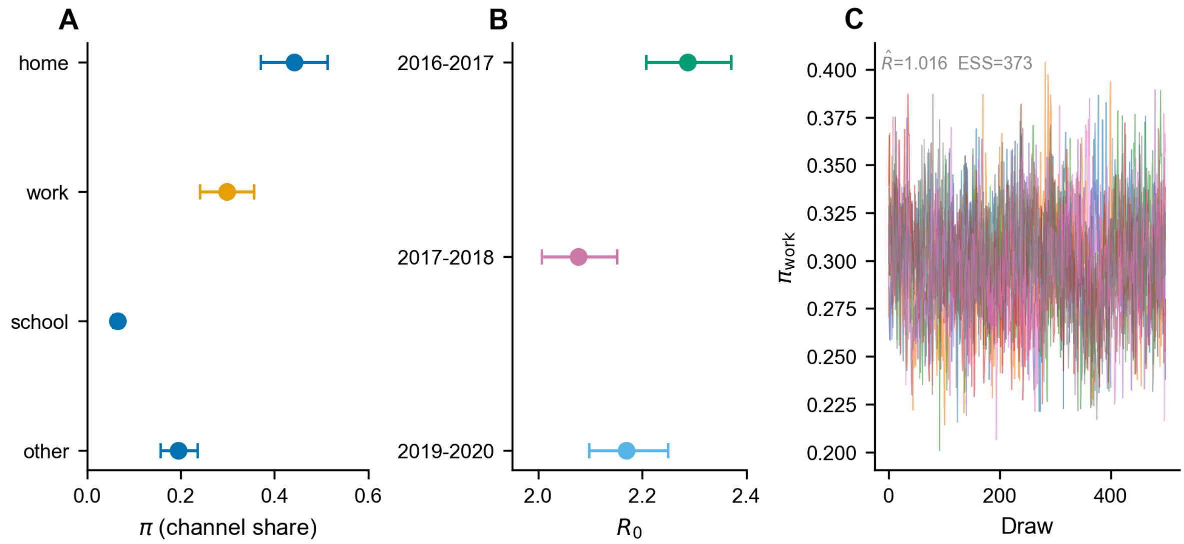

Figure 7: Three-season joint NUTS posterior for the channel allocation and season-specific reproduction numbers.

5.5 Robustness to structural assumptions and population

We stress-tested the conclusions with a sensitivity battery (six analyses over roughly 4,000 parameter combinations; Section 4.5), reporting outcomes stratified by age and by term/winter-break window rather than as a single total. Three results stand out. First, the age-redistribution direction is highly robust: sick leave lowered adult and raised child attack rate (adult-down / child-up) in all three seasons, and across the crowding sweep the child-ward winter-break transfer was positive in 749 of 750 posterior draws — the redistribution direction essentially never reverses. Second, sick leave's net effect on total infection is conditional on crowding: it is beneficial at low κ but turns net-harmful above a threshold $\kappa \approx 0.51$, which lies above the empirically derived crowding coefficient ($\kappa \approx 0.34$–0.40), so the identified regime is net-beneficial; at the low Italian workplace share the effect is harmful across all κ, and at the Ferguson-scale $\kappa = 1.0$ it is harmful across all workplace shares (Figure 8). Third, school absence remained the larger and more certain lever in every season. These conclusions held under sweeps of the presymptomatic weight w, adult initial immunity, the symmetric baseline retention p, and the Erlang shape, with no sign reversal of the redistribution direction.

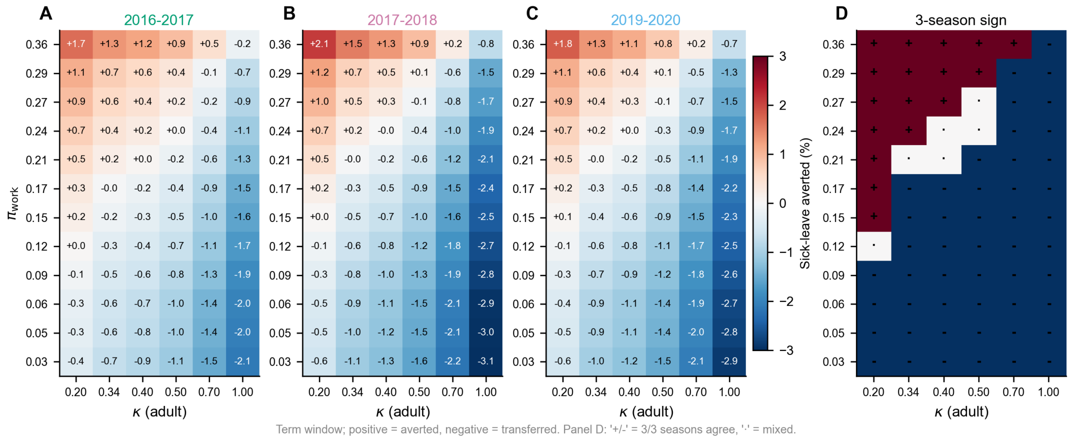

Figure 8: Sensitivity of sick leave's net effect over the workplace share π_{work} and crowding scale, by season. The sign boundary separates net-beneficial from net-harmful regimes; the empirically identified operating point lies in the net-beneficial region.

Finally, the results are robust to the population input. Re-fitting the model with year-specific resident-registration populations (June of each peak year) rather than a single reference year shifted the channel allocation by ≤ 0.004 per channel and season-specific R_0 by $< 1\%$, changed no $0 \notin$ CI status, and preserved the adult-down / child-up direction and the school-over-sick-leave ordering in all three seasons (Supplement).

6 Discussion

6.1 School absence works; sick leave is uncertain

The headline of an absenteeism analysis is usually a single averted-infection number. Our results show that the more decision-relevant contrast is one of certainty. School absence delivered a credible reduction in infection in every season and every age band; sick leave delivered a small net effect that, under the posterior, could not be distinguished from zero in two of three seasons. For a decision-maker choosing where to invest, this asymmetry — a lever that reliably works versus one whose population benefit is marginal and uncertain — is more informative than any point estimate, and it points clearly toward school-based measures.

6.2 Redistribution, metric, and timing

Beneath the aggregate, sick leave is better understood as a redistribution operator than as a reducer. It moves transmission from adults toward school-age children through the household, in a direction that reproduced across all three seasons, though by a magnitude small enough that the child increases were only partly significant — a tendency, not an established harm. Which group "bears" the redistribution depends on the metric: per-capita, children; in absolute counts, adults dominate because they are so numerous. And the redistribution toward children intensifies during the winter break, because sick-leave spillover flows through the always-open home channel rather than the school channel the holiday shuts. A policy can thus post a positive aggregate number while shifting per-capita risk onto the most vulnerable group, most strongly at the very time — the winter holidays — when families are together.

6.3 Reframing evaluation: from effect size to "who, and when"

The methodological consequence is a shift in the evaluation question. A single population-level averted number can be positive while children's per-capita burden rises, can itself be indistinguishable from zero, and can change in size across epidemic phases. For age-structured pathogens we argue that intervention appraisal should report whom the policy moves risk toward, with what certainty, and when it is applied — not only the net total, and not in a single metric. This generalises beyond the Korean setting: any absenteeism-style measure against an age-heterogeneous pathogen with presymptomatic transmission is a redistribution operator whose sign for a given group and metric is timing-dependent and often statistically fragile.

6.4 Policy implications

To anchor magnitudes, in the representative 2019–20 season school absence averted about 123,000 infections against sick leave's net ∼23,000, and sick leave's population benefit was accompanied by roughly 4,300 additional infections among children. Three implications follow. First, school-based measures are the stronger and more certain lever — larger in effect and, acting downward on the age gradient, free of the household-spillover hazard; where feasible they should be prioritized over workplace measures for influenza control. Second, sick leave's population benefit is small and not robustly established under current conditions (chiefly the empirically measured household crowding that returns removed transmission to the home), so it should not be assumed unconditionally beneficial; its specific hazard is a per-capita transfer toward children that is largest in low-workplace-transmission settings and during winter holidays. This setting-dependence reconciles an apparent tension with prior work: the low Italian workplace-contribution estimate ($\pi_{\text{work}} \approx 0.03$; Ajelli et al. [1]) would place sick leave in the net-harmful regime, whereas the Korean workplace share (≈ 0.30, consistent with domestic contact surveys [20]) sits in the beneficial regime — a difference plausibly reflecting commuting and workplace-contact patterns, and a reminder that the sign of sick leave's effect is setting-specific and should be evaluated locally rather than assumed. Third, because that hazard is a household hazard, the natural mitigation is to pair sick-leave encouragement with household-isolation support, consistent with the long-standing modelling result that home isolation must accompany, not replace, other measures [9, 5]; this is most important during school holidays.

A crude mortality overlay reinforces this ranking while underscoring its own limits. Because the age-specific fatality risk of influenza rises steeply with age — seasonal case-fatality reaches several percent in the very elderly [15] — weighting each policy's age-resolved averted infections by order-of-magnitude infection-fatality ratios (of order 10^{-5} in children rising to 10^{-2} in the elderly) widens the gap between the two levers. Under this overlay for 2019–20, school absence's averted deaths are dominated by the elderly (about two-thirds of the total), because it lowers infection in every age band including the 65+ group; sick leave yields a much smaller mortality benefit that is confined to working-age adults, and because it barely changes — and in the point estimate slightly raises — infection among the elderly, its averted deaths in that band are near zero or marginally adverse. We emphasise that this is illustrative only: the model is not designed to estimate mortality — it omits the contact, comorbidity, and care-setting structure that drive elderly influenza deaths, and the elderly infection changes on which the calculation hinges are small and statistically uncertain — so these figures should be read as a directional cross-check of the infection-based results, not as a mortality estimate.

6.5 Methodological contribution

Three methodological points may be useful to other age-structured intervention studies. The first is the explicit representation of presymptomatic transmission through an Erlang infectious period with a distinct pre-symptom sub-stage: because symptom-triggered policies cannot act on this stage, it sets a ceiling on their effectiveness — an effect invisible to models without a presymptomatic compartment. The second is the clean separation of contact frequency from intervention intensity: representing the school calendar through the contact matrix $C(t)$ rather than a transmissibility multiplier removes a baseline artefact (a fictitious 28% home-FOI inflation during the winter break) and is what makes the timing analysis possible at all. The third is the identifiability argument: workplace transmission is practically non-identifiable within a single season for a structural reason (home/work contact-matrix similarity among adults, compounded by trade-off with the presymptomatic weight) but is recovered by multi-season joint calibration, with π_{work} pinned down to $[0.24, 0.35]$ — a template for extracting channel-level parameters that any one season leaves silent.

6.6 Limitations

Several limitations temper these conclusions. Most importantly, sick leave's net benefit is statistically uncertain — its credible interval includes zero in two of three seasons — so claims about its population effect must remain guarded; the redistribution direction is consistent but its magnitude is only partly significant, and we deliberately stop short of asserting demonstrable harm to children. The winter-break intensification of that redistribution is a change in magnitude, not direction: across the sensitivity battery the child-ward transfer did not reverse sign, though its size scales with the crowding coefficient, so we frame it as "always redistributes, intensifies in the winter break." The school-versus-sick-leave effect ratio is likewise assumption-sensitive (it varies substantially with baseline symmetry, adult immunity, κ, and the presymptomatic weight), so we report only the robust direction that school absence is clearly larger. The presymptomatic weight is based on influenza A; influenza B shows no clear presymptomatic transmission [4], so applying an A-based w throughout is conservative, and we sweep w in sensitivity analysis. The crowding coefficient κ depends on definitional choices about which activities count as home time, which is why we sweep $\mu \in [0.5, 1.5]$. Workplace transmission is only practically identified, via the multi-season pin plus sensitivity, rather than freely estimated. The reporting fractions are largely US-based (with only the 15–19 band re-anchored to Korean data), presenteeism draws on general-illness data, and the model uses year-specific resident-registration populations (June of each peak year); re-fitting with these versus a single reference-year population shifted the channel allocation by ≤ 0.004 and R_0 by $< 1\%$ with no sign change (Supplement). Finally, the model over-predicts epidemic width to some degree — an intrinsic limitation of compartmental dynamics [22], most pronounced in the unusually sharp 2016–17 season — and the 0–5 band is under-fit in some seasons; the Erlang structure is used here for its presymptomatic sub-stage rather than for any material improvement in epidemic width.

6.7 Conclusion

For an age-structured pathogen with presymptomatic transmission, an absenteeism policy is a transmission-redistribution operator whose effect depends on the direction it moves risk along the age gradient, the metric used to read it, the certainty attached to it, and the time it is applied. School absence is both more effective and more certain than sick leave, and free of the household-spillover hazard; sick leave's population benefit is small and not robustly established, and it tends to move per-capita risk toward children, most strongly during winter holidays. Evaluating such policies by aggregate effect size alone can endorse a measure whose benefit is marginal while it shifts risk onto the most vulnerable at the wrong time. We suggest that "who bears the risk, with what certainty, and when" should be the primary lens for age-structured intervention appraisal, and that sick-leave measures, especially during winter holidays, be paired with household-isolation support.

Data and code availability

The influenza incidence data are individual-level insurance claims held by the Korean National Health Insurance Service (NHIS) and were accessed through the NHIS National Health Insurance Sharing Service (NHISS) research platform (`https://nhiss.nhis.or.kr`); they are available to approved researchers under NHIS data-use application and institutional review, and cannot be redistributed by the authors. Age-structured contact matrices are from the NIMS contact survey (see references). Resident-registration populations are publicly available from Statistics Korea (KOSIS, `https://kosis.kr`). The model code, calibration and forward-simulation scripts, and the analysis that generates the tables and figures are available at [repository URL to be inserted].

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